From bulk to local OER with Ta-Ni electrodes

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Towards the production of green hydrogen, alkaline water electrolysis is one of the most reassuring techniques particularly through utilizing electricity derived from renewable resources. Water oxidation half-reaction with its slow kinetics is assessed as the rate-limiting part of water splitting [1]. Therefore, development of new anode material that is economically feasible (noble metal-free), OER active and stable under harsh oxidative conditions attracts a lot of attention among the scientists during last decades. Owing to their well-defined crystal structures and chemical bonding, intermetallic compounds (IMCs) are considered as an intriguing class of materials for various electrocatalytic applications [2]. In particular, the combination of OER active Ni with other transition metals has enormous prospects. In this study, Ta-Ni compounds were investigated as OER electrocatalysts, aiming to combine the OER activity of Ni with the high corrosion resistance of Ta.

Single-phase samples of binary compounds $TaNi_3$, $TaNi_2$ and TaNi were synthesized by arc melting and subsequent annealing at 1423-1573 K. Electrochemical experiments for bulk cylindrically-shaped electrodes were carried out in 1M KOH to differentiate the OER activity of different Ta-Ni compounds. Scanning electrochemical cell microscopy (SECCM) was performed on purposely synthesized two-phase Ta-Ni samples (Figure). In SECCM, a hanging droplet of 0.1M KOH is approached to the catalyst surface, allowing nm-µm sized areas of a material to be probed in a high-throughput screening approach. In contrast to the averaged information obtained from macroscopic electrochemical measurements, SECCM enables to study OER at the local scale [3].

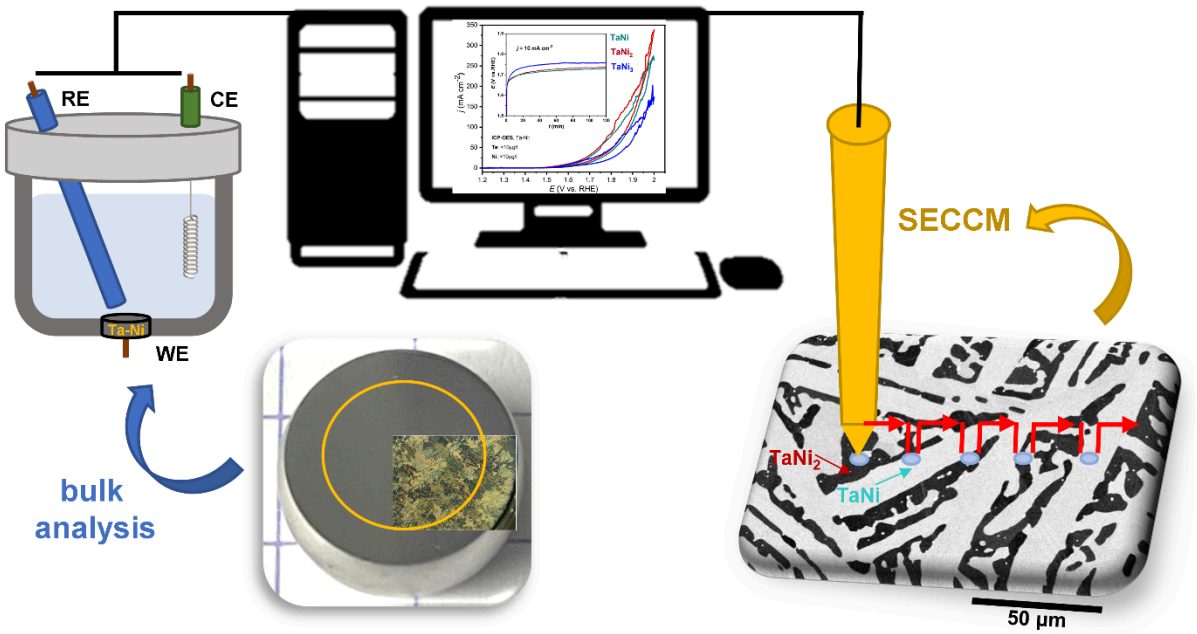

Figure. Combination of bulk electrochemical experiments with SECCM analysis on nm-µm scale using Ta-Ni electrodes.

Close OER onset overpotentials (*ca.* 370 mV) were measured for all studied Ta-Ni compounds, hence, requiring similar potentials during chronopotentiometry (CP) measurement under benchmark conditions (2h at current density of 10 mA cm^{-2}). At maximum applied potential (2.0 V vs RHE), the highest current density was achieved with $TaNi_2$ electrode, followed by TaNi and $TaNi_3$. Importantly, neither Ta nor Ni were detected in the electrolyte via ICP-OES (detection limit is 10 µg L^{-1}), revealing robustness of the studied electrode materials. In line with these results, SECCM analysis of two-phase $Ta_{37.5}Ni_{62.5}$ and $Ta_{45}Ni_{55}$ samples (consisting of $TaNi_2$ and TaNi phases in different ratios) clearly show the higher OER activity of $TaNi_2$. The electrochemical studies were accompanied by extensive characterization of the Ta-Ni electrocatalyst materials using different methods, e.g. XRD, SEM, light microscopy and XPS.

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[3] S. Schumacher, et al., *ChemElectroChem* **2022**, 9, e202200586.